

A closed-form sharp k -of- n outage bound

2026-09-20 (working note, version 0.1.0)

Abstract

Bounding the probability that at least k of n services fail *simultaneously*, given only their individual failure probabilities and no assumption about dependence, is a distribution-free extremal problem of classical Fréchet/Boole and Ruschendorf lineage. This note states and proves a *closed form* for its sharp maximum, replacing the linear program over 2^n outcome atoms that our catastrophic-outage certificate used with an exact $O(n \log n)$ sort-and-prefix-sum. We give an elementary pointwise proof of validity, an explicit attaining coupling and a matching linear-programming dual certificate for tightness, and a monotonicity corollary that turns the certificate's interval-corner argument from an assertion into a theorem. Everything is machine verified: the closed form equals the linear program to within 1.7e-16 on the worked marginals, and the certificate's outputs are byte-identical after the replacement. This is a genuine theorem of classical lineage, not a claim of complete originality.

1 Statement

Let $X_1, \dots, X_n$ be Bernoulli with marginals $p_1, \dots, p_n$ and order statistics $p_{(1)} \geq \dots \geq p_{(n)}$, and let $S = \sum_i X_i$. A *coupling* is any joint law with those marginals; nothing is assumed about dependence.

Theorem 1 (closed-form sharp k -of- n bound). *For every $1 \leq k \leq n$,*

$$\max_{\text{couplings}} \Pr(S \geq k) = \min\left(1, \min_{0 \leq j \leq k-1} \frac{1}{k-j} \sum_{i=j+1}^n p_{(i)}\right),$$

and the maximum is attained. The inner minimand at j sums the $n-j$ smallest marginals. It equals the union bound $\min(1, \sum_i p_i)$ at $k=1$ and $\min_i p_i$ at $k=n$.

2 Proofs

Validity ($\leq$). Fix a coupling and $j \in \{0, \dots, k-1\}$. On the event $\{S \geq k\}$ the j largest-marginal variables contribute at most j , so the remaining $n-j$ variables sum to at least $k-j \geq 1$; hence, at *every* outcome,

$$\mathbf{1}[S \geq k] \leq \frac{1}{k-j} \sum_{i \in B} X_i, \quad B = \text{the } n-j \text{ smallest marginals.}$$

Taking expectations (Markov) gives $\Pr(S \geq k) \leq \frac{1}{k-j} \sum_{i \in B} p_i$; minimising over j and capping at 1 yields the bound. Equivalently this is a linear-programming *dual certificate*: $\lambda_0 = 0$ and $\lambda_i = 1/(k-j^*)$ on the $n-j^*$ smallest marginals is dual-feasible (any k -subset meets B in $\geq k-j^*$ indices) with objective equal to the bound, so weak duality reproduces $\leq$.

Table 1: Sharp k -of- n bound (closed form) versus the linear-programming oracle and the classical union and Markov bounds, on the undefended worst-corner outage marginals ($n = 4$ clinical services). Percent.

k	Closed form	LP oracle	Union	Markov
1	44.91	44.91	44.91	44.91
2	22.45	22.45	44.91	22.45
3	14.97	14.97	44.91	14.97
4	9.89	9.89	44.91	11.23

Attainment ($\geq$). Let j^* be the largest minimiser and $M = \frac{1}{k-j^*} \sum_{i>j^*} p_{(i)}$; one shows $p_{(j^*+1)} \leq M \leq p_{(j^*)}$. With a uniform U on $[0, 1)$ and circumference $c = M$ (when $M < 1$): lay the $n - j^*$ smallest arcs consecutively around $[0, c)$, wrapping — total length $(k - j^*)c$ and each $\leq c$, so the covering depth is exactly $k - j^*$ throughout $[0, c)$; and let each of the j^* largest ($p_{(i)} \geq c$) cover $[0, c)$ fully. Then exactly k events fire on $[0, c)$ and fewer than k elsewhere, so $\Pr(S \geq k) = c = M$. When $M \geq 1$, laying all n arcs around the unit circle gives depth $\geq k$ everywhere, so $\Pr(S \geq k) = 1$. The full argument, and the explicit coupling as atom masses, are in `study/KOFN_THEOREM.md` and `grrc.joint_bounds.attaining_coupling`.

3 Corollaries and worked example

The bound is monotone non-decreasing in each p_i (each suffix sum is, and a capped min of non-decreasing functions is), so its worst case over an interval box of marginals sits at the all-high corner — which is exactly why the catastrophic certificate evaluates only two corners. It is non-increasing in k and never exceeds the Markov ($\sum p_i/k$) or union ($\sum p_i$) bounds, strictly below the union for $k \geq 2$ unless all marginals are equal.

Table 1 shows this on the real ATT&CK-hospital (v17.1) undefended worst-corner per-service outage marginals over $n = 4$ clinical services. The closed form matches the linear program to the last printed digit, while collapsing the “at least one” figure of 44.9% to 22.5% for two services, 15.0% for three, and 9.9% for all four — the last strictly below the Markov bound 11.2% as well.

4 Scope

The extremal problem is classical (Fréchet 1935/Boole; Ruschendorf 1991; reliability m -of- n bounds). The contribution is a clean closed form with a self-contained proof of both directions and an explicit attaining coupling, machine verified, that removes a solver from the certificate and upgrades its corner argument to a proof. The marginals themselves are evidence-anchored worst-corner model values, not measured incident frequencies.

Reproducibility. All numbers are generated from committed tables; the analysis is deterministic and carries a provenance manifest over the pinned ATT&CK bundle. The theorem is verified in `tests/test_joint_bounds.py` (closed form = LP over randomised cases, validity, attainment, duality, monotonicity, boundaries).